INCH-POUND

MIL-C-17/203
w/AMENDMENT 2
16 October 2008
SUPERSEDING
AMENDMENT 1
23 May 2002

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, TWINAXIAL, EMP HARDENED, DOUBLE SHIELD, DATA BUS, 77 OHMS

This specification is approved for use by all Departments and Agencies of the department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification sheet and MIL-DTL-17.

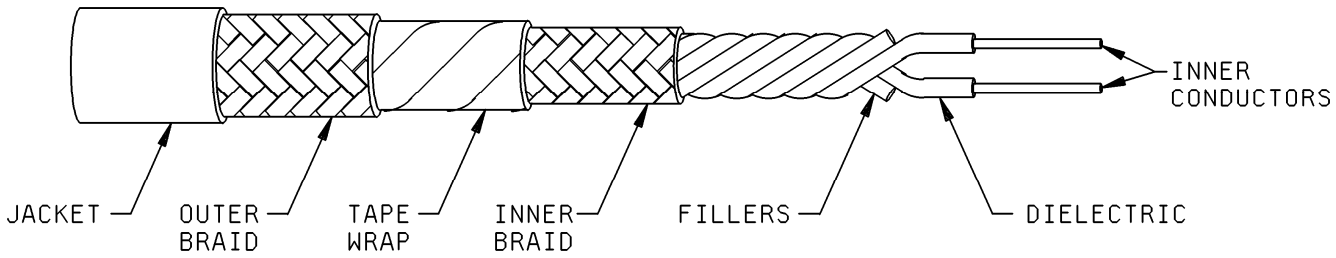


FIGURE 1. Configuration for Part or Identifying Number (PIN) M17/203-00001.

AMSC N/A

FSC 6145

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TABLE I. Description.

Components	Construction
Inner conductors 1/ 2/	M17/202-00001 AWG. 24 19 strands AWG 36 silver-coated, high strength copper alloy. O.D.: .025 ±.0001 inch.
Dielectrics 3/	Radiation cross-linked modified ETFE, one light blue, one white. Permitivity: 2.7 nominal. O.D.: .048 ±.002 inch each.
Fillers	Radiation crosslinked modified ETFE. O.D. .032 ±.002 inch.
Inner braid 4/	AWG 38 tin-coated copper. Coverage: 90% minimum. Braid angle: 18º-40º. O.D.: .114 ±.006 inch.
Tape wrap	Mumetal tape, .002 inch thick nominal; 25% overlap maximum. O.D. .130 inch maximum.
Outer braid 4/	AWG 38 tin-coated copper. Coverage: 90% minimum. Braid angle: 18º-40º O.D. .147 inch maximum.
Jacket	Radiation cross-linked Modified ETFE, white. O.D.: .161 ±.001 inch.

1/ Lay length shall be 1.00 ±.25 inch.
2/ High strength copper alloy in accordance with SAE-AS22759/35.
3/ Radiation crosslinked modified ETFE copolymer in accordance with SAE-AS22759/35.
4/ Minimum coverage of 90% in accordance with NEMA-WC27500 (shield type I, tin-coated copper).

ENGINEERING INFORMATION:

Continuous working voltage: 600 V rms, maximum, at sea level.

Operating frequency: 10 MHz, maximum.

Velocity of propagation: 61 percent, nominal.

Operating temperature range: -65°C to +150°C.

Inner conductor properties

DC resistance (maximum, at 20°C): 28.4 ohms/100 feet.

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Elongation: 6% minimum.

Tensile strength: 22.4 pounds minimum.

Jacket properties:

Elongation: 50% minimum.

Tensile strength: 5k lbf/inch^2, minimum.

Engineering note: Cables shall be suitable for use in MIL-STD-1553 data bus systems for use as main bus or stub cable.

REQUIREMENTS:

Dimensions, configuration and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors: Dielectrics shall be free stripping, using conventional stripping tools, without breaking of conductor strands and without bunching of the dielectric.

Electrical and mechanical:

Operational:

Continuity: Applicable.

Spark test: 1,000 V rms, minimum.

Voltage withstanding: 1,000 V rms, +10%, -0%.

Insulation resistance: 5,000 megohms minimum per 1,000 feet.

Corona extinction voltage: Not applicable.

Characteristic impedance: 77 ±5 ohms at 1 MHz.

Surface transfer impedance: In accordance with MIL-C-85485; 1.0 milliohms/meter maximum at 1 MHz.

RF transmission loss (attenuation): 1.4 dB/100 feet maximum at 1 MHz.

Standing wave ratio (return loss): Not applicable.

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Capacitance: 30 pF per foot, maximum at 1 MHz.

Capacitance stability: Not applicable.

Capacitance unbalance: 5% maximum.

Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

Aging stability: Not applicable.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend: -65ºC (6 inch mandrel). Jacket shall pass spark test.

Dimensional stability: Not applicable.

Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Cable shall be tested in accordance with the flammability procedure of SAE-AS22759/35 and shall meet the requirements specified herein.

Flame propagation: Not applicable.

Acid gas generation: Not applicable.

Halogen: Not applicable.

Fluid immersion: Cable shall be tested in accordance with SAE-AS22759/35 using a 6 inch diameter mandrel. There shall be no cracking of the jackets or evidence of breakdown. The increase in diameter shall be 5% maximum.

Smoke index: Not applicable.

Toxicity index: Not applicable.

Durometer hardness: Not applicable.

Weathering: Not applicable.

Abrasion resistance: Not applicable.

Tear strength: Not applicable.

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Heat distortion: Not applicable.

Physical properties of insulation: Shall be in accordance with SAE-AS22759/35.

Hot oil immersion: Not applicable.

Accelerated aging: Shall be in accordance with SAE-AS22759/35 (.5 inch mandrel, .375 pound load).

Impulse dielectric: Shall be in accordance with SAE-AS22759/35.

PINs: M17/203-00001.

Weight: 29.1 pounds/1,000 feet maximum.

Workmanship: Applicable.

Supersession: This specification shall supersede Air Force drawing 8421528 when a QPL source becomes available.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-C-85485
MIL-STD-1553
SAE-AS22759/35
NEMA-WC2700

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 6145-2007-006)

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH, TD
Air Force – 19, 99

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